

Assignment 7 Graphical User Interface

Shuifan Wang shepherd.wang@foxmail.com

May 29, 2025

GUI

This program is designed as a simplified version inspired by *GeoGebra*. It features a clean interface while offering powerful functionality, integrating CAS (Computer Algebra System), 2D graphing, and 3D calculator capabilities. Input can be made either via keyboard typing or touch-based buttons, supporting both interaction modes. Upon launching the program, it opens in CAS mode by default. You can switch between modes using the menu in the upper right corner—each mode is functionally distinct.



Figure 1: Demo 1

CAS Mode

In CAS mode, you can perform general computations and simplify single-variable expressions (by pressing the **ans** button). For example:

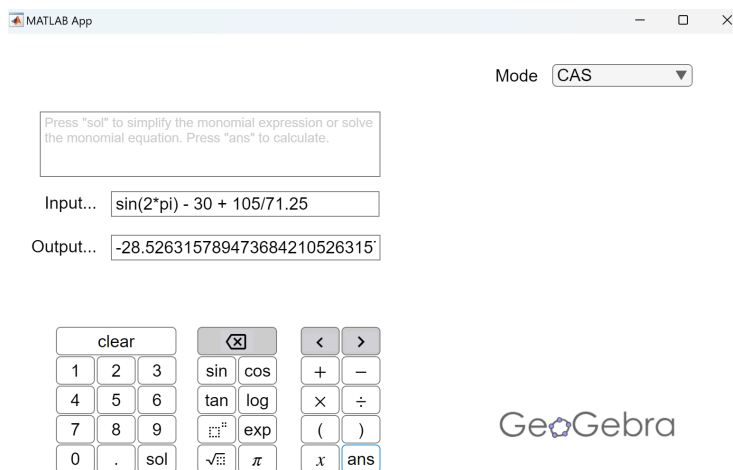


Figure 2: Demo 2

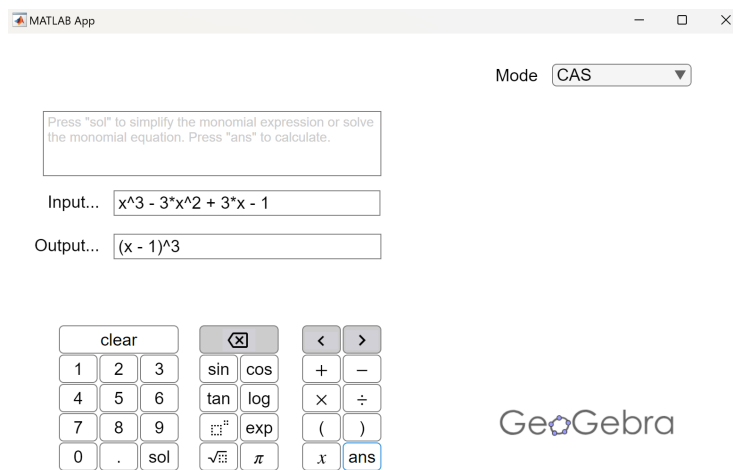


Figure 3: Demo 3

You can also solve a single-variable equation by pressing the **sol** button:

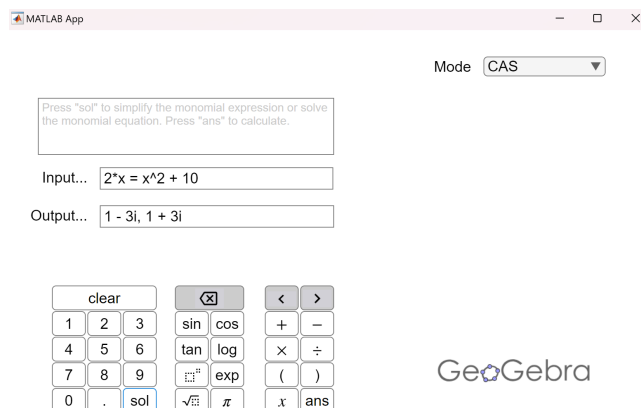


Figure 4: Demo 4

To clear all inputs instantly, simply press the **clear** button.

2D Graphing Mode

In the graphing interface, you can press **add** to include multiple input fields—this is very useful when comparing several function plots simultaneously. If you switch to another mode, remember to press **delete** to remove extra fields! You can also remove any individual field during plotting by clicking **delete**.

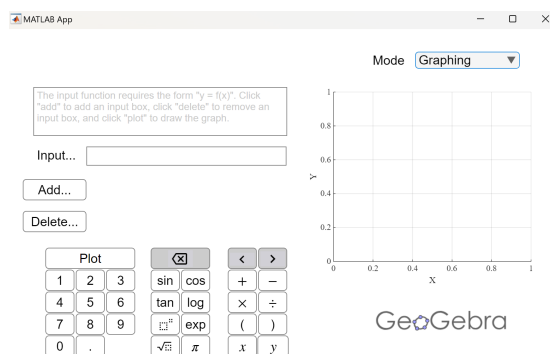


Figure 5: Demo 5

The required input format for 2D graphing is of the form $y = f(x)$:

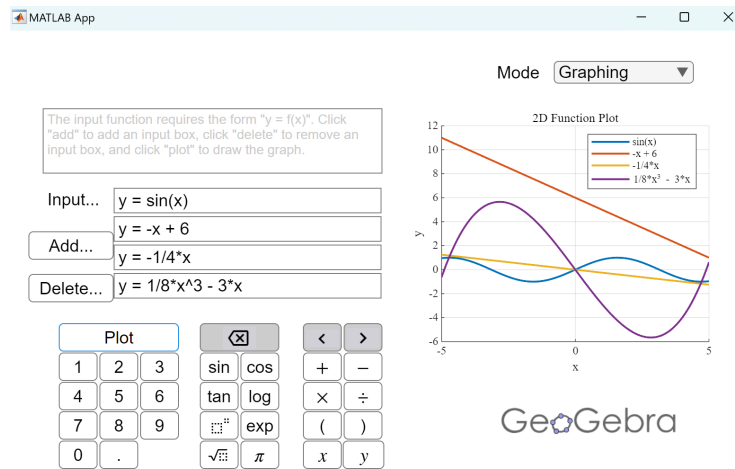


Figure 6: Demo 6

3D Calculator Mode

For 3D plotting, you need to enter functions in the format $z = g(x, y)$:

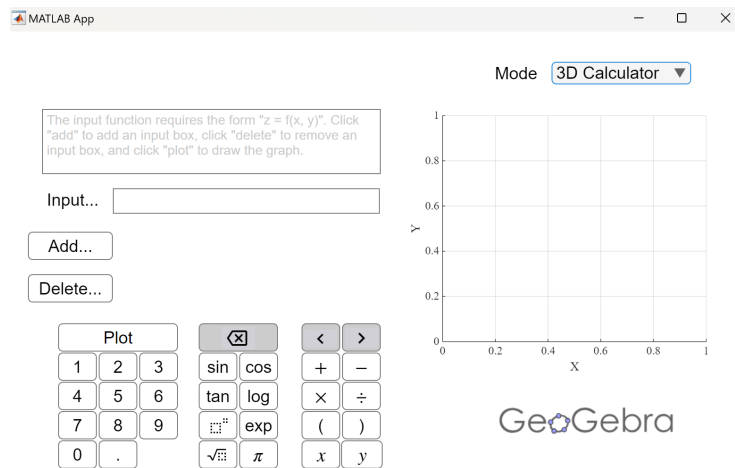


Figure 7: Demo 7

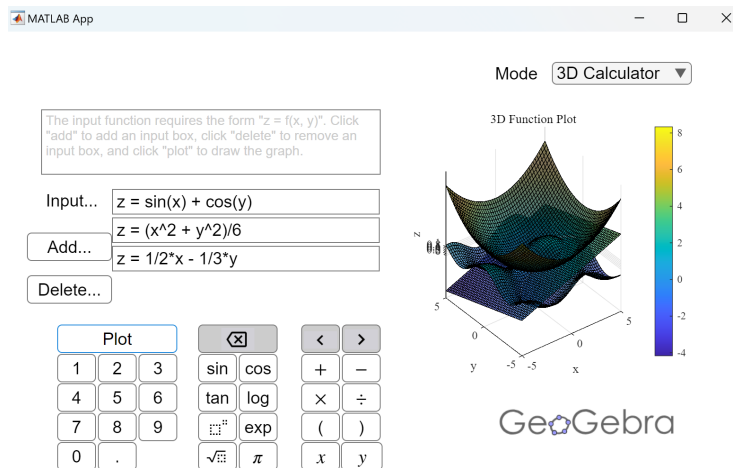


Figure 8: Demo 8

While these input requirement imposes some restrictions on user input flexibility, it improves the robustness and stability of the program to a certain extent.

Although this small app still contains many bugs and lacks some polish in functionality, I want to take this opportunity to pay tribute to the developers of free and open mathematical tools 🙌.

This program is not intended for commercial use. If any copyright is violated, please contact me at shepherd.wang@foxmail.com.

```

1  classdef GeoGebraApp < matlab.apps.AppBase
2
3  % Properties that correspond to app components
4  properties (Access = public)
5      UIFigure                matlab.ui.Figure
6      TextArea_Graphing_3D    matlab.ui.control.TextArea
7      TextArea_Graphing       matlab.ui.control.TextArea
8      PlotButton3D            matlab.ui.control.Button
9      DeleteButton3D          matlab.ui.control.Button
10     DeleteButton2D           matlab.ui.control.Button
11     Image                    matlab.ui.control.Image
12     ClearButton              matlab.ui.control.Button
13     DeleteButton             matlab.ui.control.Button
14     RightButton              matlab.ui.control.Button
15     LeftButton               matlab.ui.control.Button
16     TextArea_CAS             matlab.ui.control.TextArea
17     Button_y                 matlab.ui.control.Button
18     Button_x                 matlab.ui.control.Button
19     Button_RB                matlab.ui.control.Button
20     Button_LB                matlab.ui.control.Button
21     DivButton                matlab.ui.control.Button
22     MinusButton              matlab.ui.control.Button
23     TimesButton              matlab.ui.control.Button
24     PlusButton               matlab.ui.control.Button
25     piButton                 matlab.ui.control.Button
26     Button_dot               matlab.ui.control.Button
27     Button_0                 matlab.ui.control.Button
28     Button_9                 matlab.ui.control.Button
29     Button_6                 matlab.ui.control.Button
30     Button_3                 matlab.ui.control.Button
31     Button_8                 matlab.ui.control.Button
32     Button_5                 matlab.ui.control.Button
33     Button_2                 matlab.ui.control.Button
34     Button_7                 matlab.ui.control.Button
35     Button_4                 matlab.ui.control.Button
36     Button_1                 matlab.ui.control.Button
37     AddButton3D              matlab.ui.control.Button
38     AddButton2D              matlab.ui.control.Button
39     SolveButton              matlab.ui.control.Button
40     ModeDropDown             matlab.ui.control.DropDown
41     ModeDropDownLabel        matlab.ui.control.Label
42     CalculateButton           matlab.ui.control.Button
43     PlotButton               matlab.ui.control.Button
44     ResultOutput              matlab.ui.control.EditField
45     OutputLabel              matlab.ui.control.Label
46     ExpressionInput           matlab.ui.control.EditField
47     InputLabel               matlab.ui.control.Label
48     ExpButton                matlab.ui.control.Button
49     LogButton                matlab.ui.control.Button
50     TanButton                matlab.ui.control.Button
51     CosButton                matlab.ui.control.Button
52     SinButton                matlab.ui.control.Button
53     SqrtButton               matlab.ui.control.Button
54     PowerButton              matlab.ui.control.Button
55     UIAxes3D                 matlab.ui.control.UIAxes
56     UIAxes                   matlab.ui.control.UIAxes
57  end

```

```

58
59 % Properties for storing input fields and default plotting ...
    settings
60 properties (Access = private)
61     GraphFields matlab.ui.control.EditField = ...
        matlab.ui.control.EditField.empty; % Store ...
        input fields in 2D mode
62     Graph3DFields matlab.ui.control.EditField = ...
        matlab.ui.control.EditField.empty; % Store input ...
        fields in 3D mode
63
64     XRange = linspace(-5, 5, 1000); % Default x range
65     XMin = -5; % Default x minimum
66     XMax = 5; % Default x maximum
67     YMin = -5; % Default y minimum
68     YMax = 5; % Default y maximum
69     GridSize = 50; % Default grid size
70 end
71
72 methods (Access = private)
73
74     function updateGraph(app)
75         % Try to update the plot; on error, just clear ...
76         % the axes
77         try
78             app.drawAllFunctions();
79         catch
80             cla(app.UIAxes);
81             legend(app.UIAxes, 'off');
82         end
83     end
84
85     function drawAllFunctions(app)
86         % Clear axes and prepare for plotting
87         cla(app.UIAxes);
88         hold(app.UIAxes, 'on');
89
90         % Combine the main input and any additional fields
91         allFields = [app.ExpressionInput, app.GraphFields];
92
93         % Check if there's any valid input
94         hasValidInput = false;
95         for i = 1:numel(allFields)
96             if ~isempty(allFields(i).Value)
97                 hasValidInput = true;
98                 break;
99             end
100         end
101
102         if ~hasValidInput
103             hold(app.UIAxes, 'off');
104             return;
105         end
106
107         % Get a set of colors for plotting
108         colors = lines(numel(allFields));

```

```

109 % Process each input field and plot if valid
110 for i = 1:numel(allFields)
111     inputStr = allFields(i).Value;
112
113     % Skip empty input
114     if isempty(inputStr)
115         continue;
116     end
117
118     % Extract the expression part after '=' if ...
119         present
120     if contains(inputStr, '=')
121         exprParts = strsplit(inputStr, '=');
122         funcStr = strtrim(exprParts{2});
123     else
124         funcStr = strtrim(inputStr);
125     end
126
127     % Remove trailing semicolon if any
128     if endsWith(funcStr, ';')
129         funcStr = funcStr(1:end-1);
130     end
131
132     % Safely compute function values
133     x = app.XRange;
134     try
135         % Create anonymous function and compute y ...
136         values
137         fh = str2func(['@(x) ' vectorize(funcStr)]);
138         y = fh(x);
139
140         % Plot the curve
141         plot(app.UIAxes, x, y, ...
142             'LineWidth', 1.5, ...
143             'Color', colors(i,:), ...
144             'DisplayName', funcStr);
145     catch ME
146         % Error handling (could add error ...
147         messages here)
148     end
149
150     end
151
152     % Set axes properties
153     hold(app.UIAxes, 'off');
154     grid(app.UIAxes, 'on');
155     title(app.UIAxes, '2D Function Plot');
156     xlabel(app.UIAxes, 'x');
157     ylabel(app.UIAxes, 'y');
158 end
159
160 function updateGraph3D(app)
161 % Try to update the 3D plot
162 try
163     app.drawAll3DFunctions();
164 catch ME
165     % Display error message
166     disp(ME.message);

```



```

163         cla(app.UIAxes3D);
164     end
165 end
166
167 function drawAll3DFunctions(app)
168     % Clear the 3D axes
169     cla(app.UIAxes3D);
170
171     % Combine the main input and any additional 3D ...
172     fields
173     allFields = [app.ExpressionInput, ...
174                 app.Graph3DFields];
175
176     % Check if there's any valid input
177     hasValidInput = false;
178     for i = 1:numel(allFields)
179         if ~isempty(allFields(i).Value)
180             hasValidInput = true;
181             break;
182         end
183     end
184
185     if ~hasValidInput
186         return;
187     end
188
189     % Create a mesh grid for plotting
190     x = linspace(app.XMin, app.XMax, app.GridSize);
191     y = linspace(app.YMin, app.YMax, app.GridSize);
192     [X, Y] = meshgrid(x, y);
193
194     % Get a set of colors for multiple surfaces
195     colors = lines(numel(allFields));
196
197     % Process each input field and plot if valid
198     for i = 1:numel(allFields)
199         inputStr = allFields(i).Value;
200
201         % Skip empty input
202         if isempty(inputStr)
203             continue;
204         end
205
206         % Extract the expression part after '=' if ...
207         present
208         if contains(inputStr, '=')
209             exprParts = strsplit(inputStr, '=');
210             funcStr = strtrim(exprParts{2});
211         else
212             funcStr = strtrim(inputStr);
213         end
214
215         % Remove trailing semicolon if any
216         if endsWith(funcStr, ';')
217             funcStr = funcStr(1:end-1);
218         end
219     end
220 end

```

```

217         % Safely compute function values on the grid
218         try
219             % Create anonymous function and compute z ...
                values
220             fh = str2func(['@(x,y) ' ...
                vectorize(funcStr)]);
221             Z = fh(X, Y);
222
223             % Plot the surface
224             surf(app.UIAxes3D, X, Y, Z, 'FaceAlpha', ...
                0.7);
225             hold(app.UIAxes3D, 'on');
226
227         catch ME
228             % Error handling (display message)
229             disp(['Error in function: ' funcStr]);
230             disp(ME.message);
231         end
232     end
233
234     % Set axes properties for the 3D plot
235     hold(app.UIAxes3D, 'off');
236     grid(app.UIAxes3D, 'on');
237     title(app.UIAxes3D, '3D Function Plot');
238     xlabel(app.UIAxes3D, 'x');
239     ylabel(app.UIAxes3D, 'y');
240     zlabel(app.UIAxes3D, 'z');
241     colorbar(app.UIAxes3D);
242
243     % Add lighting for better visualization
244     light(app.UIAxes3D, 'Position', [1 1 1], 'Style', ...
        'infinite');
245     lighting(app.UIAxes3D, 'gouraud');
246     material(app.UIAxes3D, 'dull');
247
248     % Set view angle
249     view(app.UIAxes3D, 3);
250
251     end
252
253 % Callbacks that handle component events
254 methods (Access = private)
255
256     % Code that executes after component creation
257     function startupFcn(app)
258         app.ModeDropDown.Value = 'CAS';
259         app.ModeDropDown.ValueChanged(struct('Value', 'CAS'));
260     end
261
262     % Button pushed function: SinButton
263     function SinButtonPushed(app, event)
264         app.ExpressionInput.Value = ...
            [app.ExpressionInput.Value, 'sin('];
265     end
266
267     % Button pushed function: CosButton
268     function CosButtonPushed(app, event)

```

```

269         app.ExpressionInput.Value = ...
270             [app.ExpressionInput.Value , 'cos( '];
271     end
272     % Button pushed function: TanButton
273     function TanButtonPushed(app, event)
274         app.ExpressionInput.Value = ...
275             [app.ExpressionInput.Value , 'tan( '];
276     end
277     % Button pushed function: LogButton
278     function LogButtonPushed(app, event)
279         app.ExpressionInput.Value = ...
280             [app.ExpressionInput.Value , 'log( '];
281     end
282     % Button pushed function: ExpButton
283     function ExpButtonPushed(app, event)
284         app.ExpressionInput.Value = ...
285             [app.ExpressionInput.Value , 'exp( '];
286     end
287     % Button pushed function: SqrtButton
288     function SqrtButtonPushed(app, event)
289         app.ExpressionInput.Value = ...
290             [app.ExpressionInput.Value , 'sqrt( '];
291     end
292     % Button pushed function: PowerButton
293     function PowerButtonPushed(app, event)
294         app.ExpressionInput.Value = ...
295             [app.ExpressionInput.Value , '^'];
296     end
297     % Button pushed function: CalculateButton
298     function CalculateButtonPushed(app, event)
299         inputStr = app.ExpressionInput.Value;
300         % If the string contains "=", take the right-hand ...
301             side as the expression
302         eqIdx = strfind(inputStr, '=');
303         if ~isempty(eqIdx)
304             exprStr = strtrim(inputStr(eqIdx(end)+1:end));
305         else
306             exprStr = inputStr;
307         end
308         try
309             % Convert the input string to a symbolic expression
310             symExpr = str2sym(exprStr);
311             simplified = simplify(symExpr); % Perform ...
312                 algebraic simplification
313             app.ResultOutput.Value = char(simplified); % ...
314                 Display the result
315         catch
316             app.ResultOutput.Value = 'Invalid';
317         end
318     end
319 end

```

```

317 % Button pushed function: PlotButton
318 function PlotButtonPushed(app, event)
319     app.drawAllFunctions();
320 end
321
322 % Value changed function: ModeDropDown
323 function ModeDropDownValueChanged(app, event)
324     mode = app.ModeDropDown.Value;
325     switch mode
326     case 'CAS'
327         % Clear all the plots
328         cla(app.UIAxes);
329         cla(app.UIAxes3D);
330         app.UIAxes.Visible = 'off';
331         app.UIAxes3D.Visible = 'off';
332         app.ExpressionInput.Visible = 'on';
333         app.ResultOutput.Visible = 'on';
334         app.CalculateButton.Visible = 'on';
335         app.SolveButton.Visible = 'on';
336         app.Button_y.Visible = 'off';
337         app.PlotButton.Visible = 'off';
338         app.AddButton2D.Visible = 'off';
339         app.AddButton3D.Visible = 'off';
340         app.TextArea_CAS.Visible = 'on';
341         app.TextArea_Graphing.Visible = 'off';
342         app.TextArea_Graphing_3D.Visible = 'off';
343         app.ClearButton.Visible = 'on';
344         app.OutputLabel.Visible = 'on';
345         app.DeleteButton2D.Visible = 'off';
346         app.DeleteButton3D.Visible = 'off';
347         app.PlotButton3D.Visible = 'off';
348
349     case 'Graphing'
350         % Clear 2D plot
351         cla(app.UIAxes3D);
352         app.UIAxes.Visible = 'on';
353         app.UIAxes3D.Visible = 'off';
354         app.ExpressionInput.Visible = 'on';
355         app.ResultOutput.Visible = 'off';
356         app.CalculateButton.Visible = 'off';
357         app.SolveButton.Visible = 'off';
358         app.Button_y.Visible = 'on';
359         app.PlotButton.Visible = 'on';
360         app.AddButton2D.Visible = 'on';
361         app.AddButton3D.Visible = 'off';
362         app.TextArea_CAS.Visible = 'off';
363         app.TextArea_Graphing.Visible = 'on';
364         app.TextArea_Graphing_3D.Visible = 'off';
365         app.ClearButton.Visible = 'off';
366         app.OutputLabel.Visible = 'off';
367         app.DeleteButton2D.Visible = 'on';
368         app.DeleteButton3D.Visible = 'off';
369         app.PlotButton3D.Visible = 'off';
370         % Force update the 2D plot
371         app.updateGraph();
372
373     case '3D Calculator'

```

```

374         % Clear 3D plot
375         cla(app.UIAxes);
376         app.UIAxes.Visible = 'off';
377         app.UIAxes3D.Visible = 'on';
378         app.ExpressionInput.Visible = 'on';
379         app.ResultOutput.Visible = 'off';
380         app.CalculateButton.Visible = 'off';
381         app.SolveButton.Visible = 'off';
382         app.Button_y.Visible = 'on';
383         app.PlotButton.Visible = 'off';
384         app.AddButton2D.Visible = 'off';
385         app.AddButton3D.Visible = 'on';
386         app.TextArea_CAS.Visible = 'off';
387         app.TextArea_Graphing.Visible = 'off';
388         app.TextArea_Graphing_3D.Visible = 'on';
389         app.ClearButton.Visible = 'off';
390         app.OutputLabel.Visible = 'off';
391         app.DeleteButton2D.Visible = 'off';
392         app.DeleteButton3D.Visible = 'on';
393         app.PlotButton3D.Visible = 'on';
394         % Force update the 3D plot
395         app.updateGraph3D();
396     end
397 end
398
399 % Button pushed function: SolveButton
400 function SolveButtonPushed(app, event)
401     expr = app.ExpressionInput.Value;           % ...
402     % Expression input
403     symEq = str2sym(expr);
404     sol = solve(symEq, symvar(symEq));           % The ...
405     % solution
406     app.ResultOutput.Value = join(string(sol), ", ");
407 end
408
409 % Button pushed function: AddButton2D
410 function AddButton2DPushed(app, event)
411     newField = uicontrol(app.UIFigure, 'text', 'Value', ...
412         '');
413     newField.Position = [115, 284 - ...
414         30*numel(app.GraphFields), 283, 27];
415     newField.FontSize = 18;
416     newField.ValueChangedFcn = @(src, ev) app.updateGraph();
417     app.GraphFields = [app.GraphFields, newField];
418     app.updateGraph();
419 end
420
421 % Button pushed function: AddButton3D
422 function AddButton3DPushed(app, event)
423     newField = uicontrol(app.UIFigure, 'text', 'Value', ...
424         '');
425     newField.Position = [115, 284 - ...
426         30*numel(app.Graph3DFields), 283, 27];
427     newField.FontSize = 18;
428     newField.ValueChangedFcn = @(src, ev) ...
429         app.updateGraph3D();
430     app.Graph3DFields = [app.Graph3DFields, newField];

```

```

424         app.updateGraph3D();
425     end
426
427     % Button pushed function: Button_1
428     function Button_1Pushed(app, event)
429         app.ExpressionInput.Value = ...
430             [app.ExpressionInput.Value, '1'];
431     end
432
433     % Button pushed function: Button_2
434     function Button_2Pushed(app, event)
435         app.ExpressionInput.Value = ...
436             [app.ExpressionInput.Value, '2'];
437     end
438
439     % Button pushed function: Button_3
440     function Button_3Pushed(app, event)
441         app.ExpressionInput.Value = ...
442             [app.ExpressionInput.Value, '3'];
443     end
444
445     % Button pushed function: Button_5
446     function Button_5Pushed(app, event)
447         app.ExpressionInput.Value = ...
448             [app.ExpressionInput.Value, '5'];
449     end
450
451     % Button pushed function: Button_9
452     function Button_9Pushed(app, event)
453         app.ExpressionInput.Value = ...
454             [app.ExpressionInput.Value, '9'];
455     end
456
457     % Button pushed function: Button_4
458     function Button_4ButtonPushed(app, event)
459         app.ExpressionInput.Value = ...
460             [app.ExpressionInput.Value, '4'];
461     end
462
463     % Button pushed function: Button_8
464     function Button_8Pushed(app, event)
465         app.ExpressionInput.Value = ...
466             [app.ExpressionInput.Value, '8'];
467     end
468
469     % Button pushed function: Button_7
470     function Button_7Pushed(app, event)
471         app.ExpressionInput.Value = ...
472             [app.ExpressionInput.Value, '7'];
473     end
474
475     % Button pushed function: Button_6
476     function Button_6Pushed(app, event)
477         app.ExpressionInput.Value = ...
478             [app.ExpressionInput.Value, '6'];
479     end

```

```

472 % Button pushed function: Button_0
473 function Button_0Pushed(app, event)
474     app.ExpressionInput.Value = ...
        [app.ExpressionInput.Value, '0'];
475 end
476
477 % Button pushed function: Button_dot
478 function Button_dotPushed(app, event)
479     app.ExpressionInput.Value = ...
        [app.ExpressionInput.Value, '.'];
480 end
481
482 % Button pushed function: piButton
483 function piButtonPushed(app, event)
484     app.ExpressionInput.Value = ...
        [app.ExpressionInput.Value, 'pi'];
485 end
486
487 % Button pushed function: PlusButton
488 function PlusButtonPushed(app, event)
489     app.ExpressionInput.Value = ...
        [app.ExpressionInput.Value, '+'];
490 end
491
492 % Button pushed function: MinusButton
493 function MinusButtonPushed(app, event)
494     app.ExpressionInput.Value = ...
        [app.ExpressionInput.Value, '-'];
495 end
496
497 % Button pushed function: TimesButton
498 function TimesButtonPushed(app, event)
499     app.ExpressionInput.Value = ...
        [app.ExpressionInput.Value, '*'];
500 end
501
502 % Button pushed function: DivButton
503 function DivButtonPushed(app, event)
504     app.ExpressionInput.Value = ...
        [app.ExpressionInput.Value, '/'];
505 end
506
507 % Button pushed function: Button_LB
508 function Button_LBPushed(app, event)
509     app.ExpressionInput.Value = ...
        [app.ExpressionInput.Value, '('];
510 end
511
512 % Button pushed function: Button_RB
513 function Button_RBPushed(app, event)
514     app.ExpressionInput.Value = ...
        [app.ExpressionInput.Value, ')'];
515 end
516
517 % Button pushed function: Button_x
518 function Button_xPushed(app, event)
519     app.ExpressionInput.Value = ...

```

```

520         [app.ExpressionInput.Value , 'x'];
521     end
522     % Button pushed function: Button_y
523     function Button_yPushed(app, event)
524         app.ExpressionInput.Value = ...
525             [app.ExpressionInput.Value , 'y'];
526     end
527     % Button pushed function: ClearButton
528     function ClearButtonPushed(app, event)
529         app.ExpressionInput.Value = "";
530     end
531
532     % Button pushed function: DeleteButton
533     function DeleteButtonPushed(app, event)
534         % Read the current text (string or char)
535         str = app.ExpressionInput.Value;
536         if strlength(str) > 0
537             % Remove the last character
538             % If 'its a char array, use numel(str); if 'its a ...
539                 string, use strlength
540             newStr = extractBefore(str, strlength(str)); % ...
541                 Remove the last character
542             app.ExpressionInput.Value = newStr;
543         end
544     end
545     % Button pushed function: DeleteButton2D
546     function DeleteButton2DPushed(app, event)
547         % If there are no dynamically generated input fields, ...
548             return directly
549         if isempty(app.GraphFields)
550             return;
551         end
552         % 1. Delete the last generated input field component
553         lastField = app.GraphFields(end);
554         delete(lastField);
555         % 2. Remove this component handle from the ...
556             GraphFields array
557         app.GraphFields(end) = [];
558         % 3. Rearrange the positions of the remaining input ...
559             fields (maintain a gap of 30 pixels)
560         % Assume each input field has a height of 27, ...
561             initial y-coordinate is 284, and the gap is 30
562         for idx = 1:numel(app.GraphFields)
563             newY = 284 - 30*(idx-1);
564             app.GraphFields(idx).Position(2) = newY;
565         end
566         % 4. Call updateGraph() to refresh the graph based on ...
567             the modified list of input fields
568         app.updateGraph();
569     end

```



```

568
569 % Button pushed function: DeleteButton3D
570 function DeleteButton3DPushed(app, event)
571     % If there are no dynamically generated input fields, ...
572     % return directly
573     if isempty(app.Graph3DFields)
574         return;
575     end
576
577     % 1. Delete the last generated input field component
578     lastField = app.Graph3DFields(end);
579     delete(lastField);
580
581     % 2. Remove this component handle from the ...
582     % Graph3DFields array
583     app.Graph3DFields(end) = [];
584
585     % 3. Rearrange the positions of the remaining input ...
586     % fields (maintain a gap of 30 pixels)
587     % Assume each input field has a height of 27, ...
588     % initial y-coordinate is 284, and the gap is 30
589     for idx = 1:numel(app.Graph3DFields)
590         newY = 284 - 30*(idx-1);
591         app.Graph3DFields(idx).Position(2) = newY;
592     end
593
594     % 4. Call updateGraph3D() to refresh the graph based ...
595     % on the modified list of input fields
596     app.updateGraph3D();
597 end
598
599 % Button pushed function: PlotButton3D
600 function PlotButton3DPushed(app, event)
601     app.updateGraph3D();
602 end
603
604 % Component initialization
605 methods (Access = private)
606
607 % Create UIFigure and components
608 function createComponents(app)
609
610     % Get the file path for locating images
611     pathToMLAPP = fileparts(mfilename('fullpath'));
612
613     % Create UIFigure and hide until all components are ...
614     % created
615     app.UIFigure = uifigure('Visible', 'off');
616     app.UIFigure.Color = [1 1 1];
617     app.UIFigure.Position = [100 100 787 506];
618     app.UIFigure.Name = 'MATLAB App';
619
620     % Create UIAxes
621     app.UIAxes = uiaxes(app.UIFigure);
622     xlabel(app.UIAxes, 'X')
623     ylabel(app.UIAxes, 'Y')

```

```

619     xlabel(app.UIAxes, 'Z')
620     app.UIAxes.FontName = 'Times New Roman';
621     app.UIAxes.XGrid = 'on';
622     app.UIAxes.YGrid = 'on';
623     app.UIAxes.ZGrid = 'on';
624     app.UIAxes.Position = [428 145 321 280];
625
626     % Create UIAxes3D
627     app.UIAxes3D = uiaxes(app UIFigure);
628     xlabel(app.UIAxes3D, 'X')
629     ylabel(app.UIAxes3D, 'Y')
630     zlabel(app.UIAxes3D, 'Z')
631     app.UIAxes3D.FontName = 'Times New Roman';
632     app.UIAxes3D.ZTick = [0 0.2 0.4 0.6 0.8 1];
633     app.UIAxes3D.XGrid = 'on';
634     app.UIAxes3D.YGrid = 'on';
635     app.UIAxes3D.ZGrid = 'on';
636     app.UIAxes3D.Position = [428 145 321 280];
637
638     % Create PowerButton
639     app.PowerButton = uibutton(app UIFigure, 'push');
640     app.PowerButton.ButtonPushedFcn = ...
        createCallbackFcn(app, @PowerButtonPushed, true);
641     app.PowerButton.Icon = fullfile(pathToMLAPP, ...
        'power_icon.png');
642     app.PowerButton.BackgroundColor = [1 1 1];
643     app.PowerButton.Position = [206 75 41 30];
644     app.PowerButton.Text = '';
645
646     % Create SqrtButton
647     app.SqrtButton = uibutton(app UIFigure, 'push');
648     app.SqrtButton.ButtonPushedFcn = ...
        createCallbackFcn(app, @SqrtButtonPushed, true);
649     app.SqrtButton.Icon = fullfile(pathToMLAPP, ...
        'sqrt_icon.png');
650     app.SqrtButton.BackgroundColor = [1 1 1];
651     app.SqrtButton.Position = [206 44 41 30];
652     app.SqrtButton.Text = '';
653
654     % Create SinButton
655     app.SinButton = uibutton(app UIFigure, 'push');
656     app.SinButton.ButtonPushedFcn = ...
        createCallbackFcn(app, @SinButtonPushed, true);
657     app.SinButton.BackgroundColor = [1 1 1];
658     app.SinButton.FontSize = 18;
659     app.SinButton.Position = [206 137 41 30];
660     app.SinButton.Text = 'sin';
661
662     % Create CosButton
663     app.CosButton = uibutton(app UIFigure, 'push');
664     app.CosButton.ButtonPushedFcn = ...
        createCallbackFcn(app, @CosButtonPushed, true);
665     app.CosButton.BackgroundColor = [1 1 1];
666     app.CosButton.FontSize = 18;
667     app.CosButton.Position = [249 137 41 30];
668     app.CosButton.Text = 'cos';
669

```

```

670 % Create TanButton
671 app.TanButton = uibutton(app.UIFigure, 'push');
672 app.TanButton.ButtonPushedFcn = ...
        createCallbackFcn(app, @TanButtonPushed, true);
673 app.TanButton.BackgroundColor = [1 1 1];
674 app.TanButton.FontSize = 18;
675 app.TanButton.Position = [206 106 41 30];
676 app.TanButton.Text = 'tan';
677
678 % Create LogButton
679 app.LogButton = uibutton(app.UIFigure, 'push');
680 app.LogButton.ButtonPushedFcn = ...
        createCallbackFcn(app, @LogButtonPushed, true);
681 app.LogButton.BackgroundColor = [1 1 1];
682 app.LogButton.FontSize = 18;
683 app.LogButton.Position = [249 106 41 30];
684 app.LogButton.Text = 'log';
685
686 % Create ExpButton
687 app.ExpButton = uibutton(app.UIFigure, 'push');
688 app.ExpButton.ButtonPushedFcn = ...
        createCallbackFcn(app, @ExpButtonPushed, true);
689 app.ExpButton.BackgroundColor = [1 1 1];
690 app.ExpButton.FontSize = 18;
691 app.ExpButton.Position = [249 75 41 30];
692 app.ExpButton.Text = 'exp';
693
694 % Create InputLabel
695 app.InputLabel = uilabel(app.UIFigure);
696 app.InputLabel.HorizontalAlignment = 'right';
697 app.InputLabel.FontName = 'Arial';
698 app.InputLabel.FontSize = 18;
699 app.InputLabel.Position = [40 318 60 23];
700 app.InputLabel.Text = 'Input...';
701
702 % Create ExpressionInput
703 app.ExpressionInput = uieditfield(app.UIFigure, 'text');
704 app.ExpressionInput.FontSize = 18;
705 app.ExpressionInput.Position = [115 314 283 27];
706
707 % Create OutputLabel
708 app.OutputLabel = uilabel(app.UIFigure);
709 app.OutputLabel.HorizontalAlignment = 'right';
710 app.OutputLabel.FontName = 'Arial';
711 app.OutputLabel.FontSize = 18;
712 app.OutputLabel.Position = [26 272 74 23];
713 app.OutputLabel.Text = 'Output...';
714
715 % Create ResultOutput
716 app.ResultOutput = uieditfield(app.UIFigure, 'text');
717 app.ResultOutput.FontSize = 18;
718 app.ResultOutput.Position = [115 268 284 27];
719
720 % Create PlotButton
721 app.PlotButton = uibutton(app.UIFigure, 'push');
722 app.PlotButton.ButtonPushedFcn = ...
        createCallbackFcn(app, @PlotButtonPushed, true);

```

```

723 app.PlotButton.BackgroundColor = [1 1 1];
724 app.PlotButton.FontSize = 18;
725 app.PlotButton.Position = [57 168 127 30];
726 app.PlotButton.Text = 'Plot';
727
728 % Create CalculateButton
729 app.CalculateButton = uibutton(app.UIFigure, 'push');
730 app.CalculateButton.ButtonPushedFcn = ...
    createCallbackFcn(app, @CalculateButtonPushed, ...
    true);
731 app.CalculateButton.BackgroundColor = [1 1 1];
732 app.CalculateButton.FontSize = 18;
733 app.CalculateButton.Position = [358 44 41 30];
734 app.CalculateButton.Text = 'ans';
735
736 % Create ModeDropDownLabel
737 app.ModeDropDownLabel = uilabel(app.UIFigure);
738 app.ModeDropDownLabel.HorizontalAlignment = 'right';
739 app.ModeDropDownLabel.FontName = 'Arial';
740 app.ModeDropDownLabel.FontSize = 18;
741 app.ModeDropDownLabel.Position = [515 452 50 23];
742 app.ModeDropDownLabel.Text = 'Mode';
743
744 % Create ModeDropDown
745 app.ModeDropDown = uidropdown(app.UIFigure);
746 app.ModeDropDown.Items = {'CAS', 'Graphing', '3D ...
    Calculator'};
747 app.ModeDropDown.ValueChangedFcn = ...
    createCallbackFcn(app, @ModeDropDownValueChanged, ...
    true);
748 app.ModeDropDown.FontSize = 18;
749 app.ModeDropDown.Position = [580 451 148 24];
750 app.ModeDropDown.Value = 'CAS';
751
752 % Create SolveButton
753 app.SolveButton = uibutton(app.UIFigure, 'push');
754 app.SolveButton.ButtonPushedFcn = ...
    createCallbackFcn(app, @SolveButtonPushed, true);
755 app.SolveButton.BackgroundColor = [1 1 1];
756 app.SolveButton.FontSize = 18;
757 app.SolveButton.Position = [143 44 41 30];
758 app.SolveButton.Text = 'sol';
759
760 % Create AddButton2D
761 app.AddButton2D = uibutton(app.UIFigure, 'push');
762 app.AddButton2D.ButtonPushedFcn = ...
    createCallbackFcn(app, @AddButton2DPushed, true);
763 app.AddButton2D.BackgroundColor = [1 1 1];
764 app.AddButton2D.FontName = 'Arial';
765 app.AddButton2D.FontSize = 18;
766 app.AddButton2D.Position = [25 265 90 30];
767 app.AddButton2D.Text = 'Add...';
768
769 % Create AddButton3D
770 app.AddButton3D = uibutton(app.UIFigure, 'push');
771 app.AddButton3D.ButtonPushedFcn = ...
    createCallbackFcn(app, @AddButton3DPushed, true);

```

```

772 app.AddButton3D.BackgroundColor = [1 1 1];
773 app.AddButton3D.FontName = 'Arial';
774 app.AddButton3D.FontSize = 18;
775 app.AddButton3D.Position = [25 265 90 30];
776 app.AddButton3D.Text = 'Add...';
777
778 % Create Button_1
779 app.Button_1 = uibutton(app.UIFigure, 'push');
780 app.Button_1.ButtonPushedFcn = createCallbackFcn(app, ...
    @Button_1Pushed, true);
781 app.Button_1.BackgroundColor = [1 1 1];
782 app.Button_1.FontSize = 18;
783 app.Button_1.Position = [57 137 41 30];
784 app.Button_1.Text = '1';
785
786 % Create Button_4
787 app.Button_4 = uibutton(app.UIFigure, 'push');
788 app.Button_4.ButtonPushedFcn = createCallbackFcn(app, ...
    @Button_4ButtonPushed, true);
789 app.Button_4.BackgroundColor = [1 1 1];
790 app.Button_4.FontSize = 18;
791 app.Button_4.Position = [57 106 41 30];
792 app.Button_4.Text = '4';
793
794 % Create Button_7
795 app.Button_7 = uibutton(app.UIFigure, 'push');
796 app.Button_7.ButtonPushedFcn = createCallbackFcn(app, ...
    @Button_7Pushed, true);
797 app.Button_7.BackgroundColor = [1 1 1];
798 app.Button_7.FontSize = 18;
799 app.Button_7.Position = [57 75 41 30];
800 app.Button_7.Text = '7';
801
802 % Create Button_2
803 app.Button_2 = uibutton(app.UIFigure, 'push');
804 app.Button_2.ButtonPushedFcn = createCallbackFcn(app, ...
    @Button_2Pushed, true);
805 app.Button_2.BackgroundColor = [1 1 1];
806 app.Button_2.FontSize = 18;
807 app.Button_2.Position = [100 137 41 30];
808 app.Button_2.Text = '2';
809
810 % Create Button_5
811 app.Button_5 = uibutton(app.UIFigure, 'push');
812 app.Button_5.ButtonPushedFcn = createCallbackFcn(app, ...
    @Button_5Pushed, true);
813 app.Button_5.BackgroundColor = [1 1 1];
814 app.Button_5.FontSize = 18;
815 app.Button_5.Position = [100 106 41 30];
816 app.Button_5.Text = '5';
817
818 % Create Button_8
819 app.Button_8 = uibutton(app.UIFigure, 'push');
820 app.Button_8.ButtonPushedFcn = createCallbackFcn(app, ...
    @Button_8Pushed, true);
821 app.Button_8.BackgroundColor = [1 1 1];
822 app.Button_8.FontSize = 18;

```

```

823     app.Button_8.Position = [100 75 41 30];
824     app.Button_8.Text = '8';
825
826     % Create Button_3
827     app.Button_3 = uibutton(app.UIFigure, 'push');
828     app.Button_3.ButtonPushedFcn = createCallbackFcn(app, ...
        @Button_3Pushed, true);
829     app.Button_3.BackgroundColor = [1 1 1];
830     app.Button_3.FontSize = 18;
831     app.Button_3.Position = [143 137 41 30];
832     app.Button_3.Text = '3';
833
834     % Create Button_6
835     app.Button_6 = uibutton(app.UIFigure, 'push');
836     app.Button_6.ButtonPushedFcn = createCallbackFcn(app, ...
        @Button_6Pushed, true);
837     app.Button_6.BackgroundColor = [1 1 1];
838     app.Button_6.FontSize = 18;
839     app.Button_6.Position = [143 106 41 30];
840     app.Button_6.Text = '6';
841
842     % Create Button_9
843     app.Button_9 = uibutton(app.UIFigure, 'push');
844     app.Button_9.ButtonPushedFcn = createCallbackFcn(app, ...
        @Button_9Pushed, true);
845     app.Button_9.BackgroundColor = [1 1 1];
846     app.Button_9.FontSize = 18;
847     app.Button_9.Position = [143 75 41 30];
848     app.Button_9.Text = '9';
849
850     % Create Button_0
851     app.Button_0 = uibutton(app.UIFigure, 'push');
852     app.Button_0.ButtonPushedFcn = createCallbackFcn(app, ...
        @Button_0Pushed, true);
853     app.Button_0.BackgroundColor = [1 1 1];
854     app.Button_0.FontSize = 18;
855     app.Button_0.Position = [57 44 41 30];
856     app.Button_0.Text = '0';
857
858     % Create Button_dot
859     app.Button_dot = uibutton(app.UIFigure, 'push');
860     app.Button_dot.ButtonPushedFcn = ...
        createCallbackFcn(app, @Button_dotPushed, true);
861     app.Button_dot.BackgroundColor = [1 1 1];
862     app.Button_dot.FontSize = 18;
863     app.Button_dot.Position = [100 44 41 30];
864     app.Button_dot.Text = '.';
865
866     % Create piButton
867     app.piButton = uibutton(app.UIFigure, 'push');
868     app.piButton.ButtonPushedFcn = createCallbackFcn(app, ...
        @piButtonPushed, true);
869     app.piButton.BackgroundColor = [1 1 1];
870     app.piButton.FontSize = 18;
871     app.piButton.Interpreter = 'latex';
872     app.piButton.Position = [249 44 41 30];
873     app.piButton.Text = '\pi';

```

```

874
875 % Create PlusButton
876 app.PlusButton = uibutton(app UIFigure, 'push');
877 app.PlusButton.ButtonPushedFcn = ...
      createCallbackFcn(app, @PlusButtonPushed, true);
878 app.PlusButton.BackgroundColor = [1 1 1];
879 app.PlusButton.FontSize = 18;
880 app.PlusButton.Interpreter = 'latex';
881 app.PlusButton.Position = [315 137 41 30];
882 app.PlusButton.Text = '$+$';
883
884 % Create TimesButton
885 app.TimesButton = uibutton(app UIFigure, 'push');
886 app.TimesButton.ButtonPushedFcn = ...
      createCallbackFcn(app, @TimesButtonPushed, true);
887 app.TimesButton.BackgroundColor = [1 1 1];
888 app.TimesButton.FontSize = 18;
889 app.TimesButton.Interpreter = 'latex';
890 app.TimesButton.Position = [315 106 41 30];
891 app.TimesButton.Text = '\times';
892
893 % Create MinusButton
894 app.MinusButton = uibutton(app UIFigure, 'push');
895 app.MinusButton.ButtonPushedFcn = ...
      createCallbackFcn(app, @MinusButtonPushed, true);
896 app.MinusButton.BackgroundColor = [1 1 1];
897 app.MinusButton.FontSize = 18;
898 app.MinusButton.Interpreter = 'latex';
899 app.MinusButton.Position = [358 137 41 30];
900 app.MinusButton.Text = '$-$';
901
902 % Create DivButton
903 app.DivButton = uibutton(app UIFigure, 'push');
904 app.DivButton.ButtonPushedFcn = ...
      createCallbackFcn(app, @DivButtonPushed, true);
905 app.DivButton.BackgroundColor = [1 1 1];
906 app.DivButton.FontSize = 18;
907 app.DivButton.Interpreter = 'latex';
908 app.DivButton.Position = [358 106 41 30];
909 app.DivButton.Text = '\div';
910
911 % Create Button_LB
912 app.Button_LB = uibutton(app UIFigure, 'push');
913 app.Button_LB.ButtonPushedFcn = ...
      createCallbackFcn(app, @Button_LBPushed, true);
914 app.Button_LB.BackgroundColor = [1 1 1];
915 app.Button_LB.FontSize = 18;
916 app.Button_LB.Position = [315 75 41 30];
917 app.Button_LB.Text = '(';
918
919 % Create Button_RB
920 app.Button_RB = uibutton(app UIFigure, 'push');
921 app.Button_RB.ButtonPushedFcn = ...
      createCallbackFcn(app, @Button_RBPushed, true);
922 app.Button_RB.BackgroundColor = [1 1 1];
923 app.Button_RB.FontSize = 18;
924 app.Button_RB.Position = [358 75 41 30];

```

```

925     app.Button_RB.Text = ')';
926
927     % Create Button_x
928     app.Button_x = uibutton(app UIFigure, 'push');
929     app.Button_x.ButtonPushedFcn = createCallbackFcn(app, ...
930         @Button_xPushed, true);
931     app.Button_x.BackgroundColor = [1 1 1];
932     app.Button_x.FontSize = 18;
933     app.Button_x.Interpreter = 'latex';
934     app.Button_x.Position = [315 44 41 30];
935     app.Button_x.Text = '$x$';
936
937     % Create Button_y
938     app.Button_y = uibutton(app UIFigure, 'push');
939     app.Button_y.ButtonPushedFcn = createCallbackFcn(app, ...
940         @Button_yPushed, true);
941     app.Button_y.BackgroundColor = [1 1 1];
942     app.Button_y.FontSize = 18;
943     app.Button_y.Interpreter = 'latex';
944     app.Button_y.Position = [357 44 44 30];
945     app.Button_y.Text = '$y$';
946
947     % Create TextArea_CAS
948     app.TextArea_CAS = uitextarea(app UIFigure);
949     app.TextArea_CAS.FontSize = 14;
950     app.TextArea_CAS.FontColor = [0.8 0.8 0.8];
951     app.TextArea_CAS.Position = [40 356 359 69];
952     app.TextArea_CAS.Value = {'Press "sol" to simplify ...
953         the monomial expression or solve the monomial ...
954         equation. Press "ans" to calculate.'};
955
956     % Create LeftButton
957     app.LeftButton = uibutton(app UIFigure, 'push');
958     app.LeftButton.Icon = fullfile(pathToMLAPP, ...
959         'left_icon.png');
960     app.LeftButton.BackgroundColor = [0.8 0.8 0.8];
961     app.LeftButton.FontSize = 18;
962     app.LeftButton.Interpreter = 'latex';
963     app.LeftButton.Position = [315 168 41 30];
964     app.LeftButton.Text = '';
965
966     % Create RightButton
967     app.RightButton = uibutton(app UIFigure, 'push');
968     app.RightButton.Icon = fullfile(pathToMLAPP, ...
969         'right_icon.png');
970     app.RightButton.BackgroundColor = [0.8 0.8 0.8];
971     app.RightButton.FontSize = 18;
972     app.RightButton.Interpreter = 'latex';
973     app.RightButton.Position = [358 168 41 30];
974     app.RightButton.Text = '';
975
976     % Create DeleteButton
977     app.DeleteButton = uibutton(app UIFigure, 'push');
978     app.DeleteButton.ButtonPushedFcn = ...
979         createCallbackFcn(app, @DeleteButtonPushed, true);
980     app.DeleteButton.Icon = fullfile(pathToMLAPP, ...
981         'backspace_icon.png');

```



```

974 app.DeleteButton.BackgroundColor = [0.8 0.8 0.8];
975 app.DeleteButton.FontSize = 18;
976 app.DeleteButton.Interpreter = 'latex';
977 app.DeleteButton.Position = [206 168 84 30];
978 app.DeleteButton.Text = '';
979
980 % Create ClearButton
981 app.ClearButton = uibutton(app.UIFigure, 'push');
982 app.ClearButton.ButtonPushedFcn = ...
    createCallbackFcn(app, @ClearButtonPushed, true);
983 app.ClearButton.BackgroundColor = [1 1 1];
984 app.ClearButton.FontSize = 18;
985 app.ClearButton.Position = [57 168 127 30];
986 app.ClearButton.Text = 'clear';
987
988 % Create Image
989 app.Image = uiimage(app.UIFigure);
990 app.Image.Position = [508 44 220 102];
991 app.Image.ImageSource = fullfile(pathToMLAPP, ...
    'geogebra.png');
992
993 % Create DeleteButton2D
994 app.DeleteButton2D = uibutton(app.UIFigure, 'push');
995 app.DeleteButton2D.ButtonPushedFcn = ...
    createCallbackFcn(app, @DeleteButton2DPushed, true);
996 app.DeleteButton2D.BackgroundColor = [1 1 1];
997 app.DeleteButton2D.FontName = 'Arial';
998 app.DeleteButton2D.FontSize = 18;
999 app.DeleteButton2D.Position = [25 220 90 30];
1000 app.DeleteButton2D.Text = 'Delete...';
1001
1002 % Create DeleteButton3D
1003 app.DeleteButton3D = uibutton(app.UIFigure, 'push');
1004 app.DeleteButton3D.ButtonPushedFcn = ...
    createCallbackFcn(app, @DeleteButton3DPushed, true);
1005 app.DeleteButton3D.BackgroundColor = [1 1 1];
1006 app.DeleteButton3D.FontName = 'Arial';
1007 app.DeleteButton3D.FontSize = 18;
1008 app.DeleteButton3D.Position = [25 220 90 30];
1009 app.DeleteButton3D.Text = 'Delete...';
1010
1011 % Create PlotButton3D
1012 app.PlotButton3D = uibutton(app.UIFigure, 'push');
1013 app.PlotButton3D.ButtonPushedFcn = ...
    createCallbackFcn(app, @PlotButton3DPushed, true);
1014 app.PlotButton3D.BackgroundColor = [1 1 1];
1015 app.PlotButton3D.FontSize = 18;
1016 app.PlotButton3D.Position = [57 168 127 30];
1017 app.PlotButton3D.Text = 'Plot';
1018
1019 % Create TextArea_Graphing
1020 app.TextArea_Graphing = uitextarea(app.UIFigure);
1021 app.TextArea_Graphing.FontSize = 14;
1022 app.TextArea_Graphing.FontColor = [0.8 0.8 0.8];
1023 app.TextArea_Graphing.Position = [40 356 359 69];
1024 app.TextArea_Graphing.Value = {'The input function ...
    requires the form "y = f(x)". Click "add" to add ...

```

```

1025         an input box, click "delete" to remove an input ...
1026         box, and click "plot" to draw the graph.'];
1027
1028     % Create TextArea_Graphing_3D
1029     app.TextArea_Graphing_3D = uitextarea(app.UIFigure);
1030     app.TextArea_Graphing_3D.FontSize = 14;
1031     app.TextArea_Graphing_3D.FontColor = [0.8 0.8 0.8];
1032     app.TextArea_Graphing_3D.Position = [40 356 359 69];
1033     app.TextArea_Graphing_3D.Value = {'The input function ...
1034         requires the form "z = f(x, y)". Click "add" to ...
1035         add an input box, click "delete" to remove an ...
1036         input box, and click "plot" to draw the graph.'};
1037
1038     % Show the figure after all components are created
1039     app.UIFigure.Visible = 'on';
1040 end
1041
1042 % App creation and deletion
1043 methods (Access = public)
1044
1045     % Construct app
1046     function app = GeoGebraApp
1047
1048         % Create UIFigure and components
1049         createComponents(app)
1050
1051         % Register the app with App Designer
1052         registerApp(app, app.UIFigure)
1053
1054         % Execute the startup function
1055         runStartupFcn(app, @startupFcn)
1056
1057         if nargin == 0
1058             clear app
1059         end
1060     end
1061
1062     % Code that executes before app deletion
1063     function delete(app)
1064
1065         % Delete UIFigure when app is deleted
1066         delete(app.UIFigure)
1067     end
1068 end
1069 end

```